# Optimization report

What was done, why the settings were chosen, and how the 4090 target compares with the tested 3080.

The project was updated from the v0.5.0 MetaHuman release, with Gemma 4 12B Q8 as the primary dialogue model and all four existing assigned cloned voices. Models, runtimes and setup tools were packaged and uploaded to the private repository.

**Measured software gain on the 3080:** warmed appraisal-plus-dialogue preparation fell from **22.82 s to 14.51 s**, about **36% lower**. This is a small text-generation screen, not microphone-to-audible response time.

## Hardware comparison

| Component | Tested preparation PC | Prepared target PC |
|---|---|---|
| Processor | Intel Core i9-14900K | Intel Core i9-13900K |
| System RAM | 32 GB | 64 GB: twice the memory capacity |
| GPU / VRAM | RTX 3080, 10 GB variant | RTX 4090, 24 GB: 2.4x the VRAM capacity |
| Dialogue allocation | 32 GPU layers; 512/256 batches in the recorded screen | Default: 32 GPU layers; 256/128 batches |
| Speech / rendering | No speech or VR in the dialogue screen; separate NF4 voice probes | BF16 cloned speech by default; VR validation still required |

## Recorded 3080 timings

| Trial - appraisal + dialogue only | Baseline format | Optimized format |
|---|---|---|
| First recorded trial | 27.016 s | 19.106 s |
| Warm repeat 1 | 22.717 s | 14.420 s |
| Warm repeat 2 | 22.921 s | 14.607 s |
| Mean of the two warm repeats | 22.82 s | 14.51 s |

Three identical-prompt trials per mode used the same Gemma Q8 model, hardware and 32-layer offload. Warm dialogue output fell from about 195 to 113 tokens; spoken replies were roughly 27-30 versus 19-21 words. Decode throughput stayed near 10 tokens/s in both modes. Baseline warm requests already reused cached prompts, so this test does not isolate each cache optimization.

**Interpretation:** the measured gain is from the combined software/output changes on the 3080. More 4090 VRAM should provide more room for simultaneous dialogue, speech and graphics, but 2.4x memory does not mean 2.4x response speed. CPU models and RAM also differ. No 4090 latency measurement was performed.

## Sources and delivered project

Private project repository | 26.87 GB package release. Source revision f0507e8; base release v0.5.0, commit e170024. All 21 release assets were checksum verified.

Settings: services/config.rtx4090.example.json; tools/optimization-settings.py; inference_runtime.py; alex_service.py; higgs_service.py. Evidence in the package's validation/ folder: summary.json, dialogue-benchmark.json, voice-benchmark.json, mtp-smoke.json and service-tests.txt. The recorded dialogue screen used 512/256 batches; the shipped default uses 256/128.

# Default settings implemented

Pipeline: microphone > speech recognition > appraisal > dialogue > cloned voice > lip sync / playback.

| Setting | Enabled value / behavior | Purpose and tradeoff |
|---|---|---|
| Primary dialogue model | Gemma 4 12B Q8_0; about 12.7 GB weight file | 8-bit model weights reduce storage and memory use while retaining the requested model. The matching MTP draft is included. |
| GPU / CPU allocation | 32 GPU layers; remaining layers in RAM; 8 LLM CPU threads | Plans GPU headroom for BF16 speech and VR. Partial CPU offload trades some dialogue speed for memory headroom. |
| Prompt batches | Batch 256; microbatch 128 | Limits the amount of prompt work submitted at once. Smaller batches aim to reduce memory peaks and competition with rendering. |
| Attention / working cache | Flash Attention on; Q8 key/value cache | Uses a more memory-efficient attention path and 8-bit storage for cached model calculations. This cache precision is separate from model-weight precision. |
| Reusable contexts | Two slots, 4,096 tokens each; prompt caching on | Appraisal and dialogue reuse separate contexts, limiting how often one stage replaces the other stage's cached prefix. |
| Stable instructions / memory | Fixed rules first; changing state and recalled facts in latest message | Encourages prefix reuse. Recent history and recalled memory are budgeted; old complete turn pairs are trimmed while output space is reserved. |
| Compact response controls | Text, emotion, intensity, gesture, voice style and gaze | Generates six essential fields. Animation timings are calculated locally, reducing JSON output while preserving the existing delivery controls. |
| Reply / reasoning budget | Prefer 15-30 spoken words; 384 tokens normally, 600 for fuller replies; appraisal 100 | Brief meaningful answers avoid padding. Detailed replies remain possible. Length/language/schema validation remains; lengthy thinking output is disabled. |
| Cloned speech / warmup | Higgs BF16; resident model; encoded reference cache; fixed openings prepared | Preserves the selected voice precision and emotional recipes. Models and voice references are reused; authored openings are prepared before play. |
| Speech recognition / threads | Whisper large-v3-turbo on CPU, INT8, 8 threads; preloaded. Higgs CPU threads: 1 | Keeps recognition off the shared GPU and limits CPU pool contention. Serbian recognition with Latin output and a Montenegrin prompt is retained; prior-utterance conditioning is off. |
| Launch / diagnostics | Duplicate game prevention; configuration signatures; stage and GPU/RAM logs | Avoids extra GPU workload and stale service settings. Timings, token/cache counts and resource samples support tuning; performance logs omit dialogue text. |
| Alternative speech path | Kokoro model and validated voice cached; startup warmup supported | The optional Kokoro path avoids repeated loading/validation. Higgs remains the default engine for the four assigned cloned voices. |

Speaker assignments remain Ivan = Igor, Stefan = Person 1, Nikola = Person 2 and Alex = Person 3. Original references and per-emotion recipes were retained. Audio quality checks and validated retries remain active.

A token is a piece of text, not a spoken word: the 384/600 limits also include JSON controls. Speech-engine BF16 precision and Gemma/cache Q8 precision are independent settings.

# Optional profiles and evidence

Close the game, open Optimization Settings.cmd, choose a profile, then restart. Each change saves a configuration backup and stops this installation's recorded AI services.

| Menu choice | What it changes | How to interpret it |
|---|---|---|
| 1. BF16 quality | BF16 speech; 32 GPU layers; 256/128 batches; MTP off | Selected default; prioritizes the retained voice quality and planned VRAM headroom. |
| 2. NF4 comparison | 4-bit NF4 speech; all Gemma layers on GPU (99 setting); 512/256 batches; MTP off | Frees speech-model memory to aim for faster dialogue. Experimental; voice/emotion quality must be checked. |
| 3. INT8 comparison | 8-bit speech; all Gemma layers on GPU; 256/128 batches; MTP off | Another memory/quality tradeoff. Can be slower than BF16; check memory and every speaker. |
| 4. Smaller VR batches | 256/128 batches; 8 LLM threads | Reduces prompt-work peaks; keeps the current voice precision, layer allocation and MTP choice. |
| 5. Larger batches | 512/256 batches; 8 LLM threads | Prompt-throughput comparison; may increase memory pressure or competition with graphics. |
| 6. MTP on | Matching draft on; up to two proposed tokens ahead | Primary model verifies draft tokens. Adds memory/work; valid JSON was tested, but VR speed benefit is unproven. |
| 7. MTP off | Disables the speculative draft | Removes draft overhead; preserves the other current settings. |

The batch and MTP choices are incremental, not complete resets. Every choice selects Gemma Q8. Quality restores BF16 speech, 32 GPU layers, 256/128 batches and MTP off.

## Additional supported options

An external appraisal endpoint can move assessment to a separately installed helper model, including a second GPU; that helper is not bundled or benchmarked. Quest refresh rate (a 72 Hz comparison), rendering resolution and a per-application driver FPS limit are manual options on the target PC.

## What was actually verified

**174 service/helper tests passed**, all four player character/face checks passed, and Q8 plus matching MTP loaded. MTP produced valid constrained JSON; this was a smoke test, not a demonstrated real-turn speedup.

**Separate 3080 NF4 voice probes:** Alex neutral 32.358 s; Ivan angry 3.047 s; Stefan failed the audio-quality check after retry; Nikola was not tested. Cold speaker preparation affected results. These are not matched BF16/NF4 benchmarks or full conversation latency measurements.

**The <15-second audible-response target is still unverified.** On the 4090, use Benchmark Responses.cmd with VR active to collect prepared-speech median/p90 timings; an optional WAV includes recognition. Also measure microphone release to actual audible playback and frame stability. The benchmark does not play its clips.